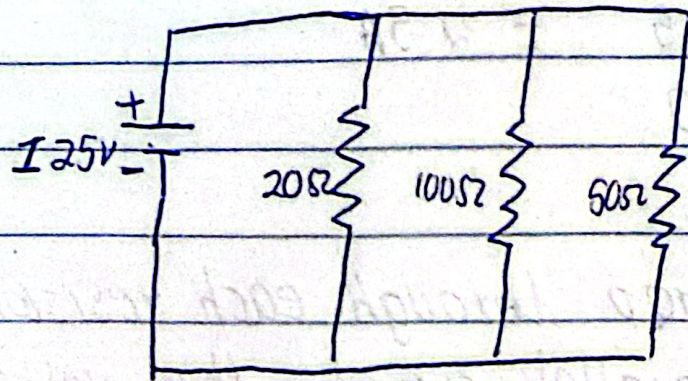




1. Total ~~current~~ resistance $\left[\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right]^{-1}$

$$\left[\frac{1}{20} + \frac{1}{100} + \frac{1}{50} \right]^{-1} = \frac{2}{25}^{-1} = \frac{25}{2} = 12.5$$

$$12.5\Omega$$

2. Total current from the power supply

$$I = \frac{V}{R} = \frac{125}{12.5} = 10A$$

$$\Rightarrow 10A$$

3. Total current through each resistor

$$I = \frac{V}{R} \quad a) \frac{125}{20} = 6.25A$$

$$b) \frac{125}{100} = 1.25A$$

$$c) \frac{125}{50} = 2.5A$$

4) Voltage drop through each resistor

For a parallel circuit, the voltage across every branch is equal to the supply voltage.

$$20\Omega \quad V_1 = 125V$$

$$100\Omega \quad V_2 = 125V$$

$$50\Omega \quad V_3 = 125V$$

5. Power dissipated by Each Resistor

$$P = V \times I$$

$$V = 125V$$

$$I = 6.25A \quad 20\Omega = 125 \times 6.25 = 781.25W$$

$$\Rightarrow 781.25W$$

$$P = V \times I$$

$$V = 125V$$

$$I = 1.25A \quad 100\Omega = 125 \times 1.25 = 156.25W$$

$$\Rightarrow 156.25W$$

$$P = V \times I$$

$$V = 125V$$

$$I = 2.5A \quad 50\Omega = 125 \times 2.5 = 312.5W$$

$$\Rightarrow 312.5W$$